DEPARTMENT OF THE AIR FORCE
Headquarters US Air Force
Washington, D.C. 20330-1030

QTP24-3-D752
25 AUGUST 2026

Excavator

Vehicle Management Codes: D752, D753, D754, D759, D798



QUALIFICATION TRAINING PACKAGE

Chapter 1—OVERVIEW **4**
1.1. Overview 4
1.2. Applicability 4
1.3. Administrative 4
Chapter 2—RESPONSIBILITIES **5**
2.1. Trainer 5
2.2. Trainee 5
2.3. Vehicle Control Official (VCO)/Unit Training Manager (UTM) 5
Chapter 3—INTRODUCTION **6**
3.1. Objectives 6
3.2. Desired Learning Outcome 6
3.3. Lesson Duration 6
Table 3.1. Recommended Training Time for Training Activities 6
3.4. Instructional References 7
3.5. Instructional Training Aids and Equipment 7
Chapter 4—KNOWLEDGE LECTURE **8**
4.1. Overview of Training and Requirements 8
Table 4.1. Control Indicators 9
Figure 4.1. Control Pattern 11
4.2. Vehicle Inspection 11
Figure 4.2. Hydraulic Fluid Indicator 12
Figure 4.3. Engine Compartment 13
Figure 4.4. Master Switch 14
Figure 4.5. Steering Wheel Controls 15
Figure 4.6. Track lever/pedal Controls 16
Figure 4.7. Operator Controls (Right) 16
Figure 4.8. Operator Controls (Left) 16
Figure 4.9. Safety Lever 17
4.3. Vehicle Safety and Equipment 18
4.4. Driving Safety and Precautions 19
4.5. Vehicle Operation 21
Figure 4.10. Attachment Installation 21
Figure 4.11. Attachment Removal 22

Figure 4.12. Ditching Work 24
Figure 4.13. Dump Truck Loading 25
Figure 4.14. Working on a Slope 26
Chapter 5—EXPLANATION AND DEMONSTRATION 29
5.1. Instructor's Preparation 29
5.2. Safety Procedures and Equipment 29
5.3. Operator Maintenance Demonstration 30
5.4. Operation Demonstration 30
Chapter 6—TRAINEE PERFORMANCE AND EVALUATION 33
6.1. Trainee Performance 33
6.2. Performance Evaluation 35
Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 39
Attachment 2—PERFORMANCE TEST 41
Attachment 3—SEVEN-STEP INSPECTION PROCESS 43

Chapter 1

OVERVIEW

1.1. Overview. This Qualification Training Package (QTP) establishes standardized training requirements and content for all Department of the Air Force (DAF) personnel responsible for the operation of the vehicle management codes listed on the cover section of this package. Designed to ensure consistency and excellence across all installations, this package provides comprehensive guidance for both trainers and trainees to ensure the safe operations of these vehicles.

1.1.1. The training standards outlined herein align with DAF requirements and supplement federal regulations. This package serves as the authoritative source for curriculum development and performance evaluation. Trainers will utilize this guide for trainee preparation, knowledge lectures, provide demonstrations, and administer a performance evaluation.

1.1.2. Trainees are required to read this entire lesson plan prior to starting lectures and must follow along with these materials during instruction. Successful completion of this training program demonstrates operator proficiency and supports mission readiness by maintaining the highest standards of safety, compliance, and operational effectiveness.

1.2. Applicability. This QTP applies to all DAF personnel tasked with operating the vehicles applicable to this package.

1.2.1. Trainees will complete a hands-on Performance Test, Attachment 2.

1.2.2. Upon successful completion, the trainer will document the results and sign off the respective tasks in the member's Individual Training Record (ITR), DAF Form 623, *Individual Training Record Folder,* or electronic On-the-Job Training () record.

1.3. Administrative. Send comments and suggested improvements on DAF Form 847, *Recommendation for Change of Publication* through Air Force Installation and Mission Support Center (AFIMSC) functional managers via e-mail at: AFIMSC.A34OG.GroundTrans@us.af.mil.

Chapter 2

RESPONSIBILITIES

2.1 Trainer. Provide an overview of the training outlined in Chapters 2 and 3. Deliver classroom lectures and conduct trainee preparation using the lesson plans. Provide knowledge overview training, conduct demonstrations, and evaluate performance activities. Coordinate with the Unit Training Manager (UTM) or designated authority to ensure training is properly documented.

2.1.1. Review this QTP and applicable manufacturer operator's manuals.

2.1.2. Conduct knowledge training with the trainees using this QTP.

2.1.3. Conduct performance task explanations and demonstration using this QTP.

2.1.4. Evaluate the trainee's task performance. The trainee must complete the performance test without assistance.

2.1.5. Document and sign-off the completed task(s) in the trainee's ITR, AF Form 623, or electronic OJT (On-the-Job Training) records.

2.2. Trainee. Read this QTP in its entirety prior to the lecture. Complete the required performance test.

2.2.1. Ensure the trainer explains the QTP process and your responsibilities. Ask questions if you do not understand the objectives, safety requirements, or principles for each section.

2.2.2. Must have in his/her possession a valid state driver's license.

2.2.3. Read DAFMAN 91-203, Chapter 30 (MOTOR VEHICLE – OPERATIONS AND MAINTENANCE)

2.2.4. Read DAFI 24-301, Chapter 1, Chapter 2, paragraph 2.13.4 Chapter 5, paragraph 5.5 and 5.6.

2.2.5. Read manufacturer's operator's manual for the vehicle being trained on.

2.2.6. Complete the Performance Test with zero instructor assists.

2.3. Vehicle Control Official (VCO)/UTM. Track the execution and completion of requirements. Ensure all testing results and task sign-off are properly loaded into the members' ITR, DAF Form 623, or electronic OJT records.

Chapter 3

INTRODUCTION

3.1. Objectives. Upon completion of lectures, demonstrations, and hands-on sessions, trainees will successfully meet all training requirements. This training ensures the trainee becomes a qualified operator who possesses the knowledge and skills to properly operate the vehicles applicable to this package. Trainees must complete all components of the performance test with zero instructor assists.

3.1.1. Train and qualify each trainee in safe operation and preventive maintenance of the excavator.

3.1.2. This training will ensure the trainee becomes a qualified excavator operator; an operator who has the knowledge and skills to operate an excavator in a safe, proficient and professional manner.

3.2. Desired Learning Outcome.

3.2.1. Understand the purpose of the excavator and its role in the mission.

3.2.2. Understand the safety precautions to be followed for pre-, during- and post-operation inspections of the excavator.

3.2.3. Know the proper operator maintenance procedures of the excavator IAW applicable technical manual(s) and use of DAF Form 1800, *Operator's Inspection Guide and Trouble Report*.

3.2.4. Be completely familiar with the safety features of the excavator.

3.2.5. Safely and proficiently operate the excavator.

3.3. Lesson Duration. Recommended instructional and hands-on training time is 26 hours. **Note:** This is a recommended baseline; training time may vary depending on how quickly a trainee demonstrates proficiency in the required tasks.

Table 3.1. Recommended Training Time for Training Activities..

Training Activity	Training Time
Trainee's Preparation	30 Minutes
Instructor's Lecture	1 Hour
Instructor's Demonstration	2.5 Hours
Trainee's Personal Experience (to build confidence and proficiency) ▪ Perform Operator Maintenance ▪ Operate the Vehicle	20 Hours
Trainee's Performance Evaluation	2 Hours

3.4. Instructional References.

3.4.1. Risk Management (RM) and Safety Principles IAW Department of the Air Force Pamphlet (DAFPAM) 90-803, *Risk Management (RM) Guidelines and Tools*.

3.4.2. Applicable technical manual(s) or Manufacturer's Operator's Manual (see Vehicle Management for technical manual(s) number for vehicle being used in training).

3.4.3. QTP 24-3-General Vehicle Safety

3.4.4. DAF Form 1800.

3.4.5 Department of the Air Force (DAFMAN) 91-203, *Department of the Air Force Occupational Safety Fire and Health Standards,*

3.4.6. DAFI 24-302, *Vehicle Management*.

3.5. Instructional Training Aids and Equipment.

3.5.1. Excavator Lesson Plan.

3.5.2. Excavator.

3.5.3. Applicable technical manual or manufacturer's operator's manual.

3.5.4. DAF Form 1800.

3.5.5. Suitable training area.

Chapter 4

KNOWLEDGE LECTURE

4.1. Overview of Training and Requirements.

4.1.1. Training objectives:

4.1.1.1. Given lectures, demonstrations, hands-on operating session(s), the trainee must be able to perform operator's inspection and complete the performance evaluation with zero instructor assists.

4.1.1.2. Train and qualify each trainee in safe operation and preventive/operational maintenance of the excavator.

4.1.1.3. This training will ensure the trainee becomes a qualified excavator operator—an operator who has the knowledge and skills to operate an excavator in a safe, proficient and professional manner.

4.1.2. Desired learning outcomes:

4.1.2.1. Understand the principals of operation, the purpose of the excavator and its role in the mission.

4.1.2.1.1. The excavator is a versatile piece of equipment primarily used for digging and moving earth. With various attachments, it can also perform tasks similar to other machinery, making it a multi-purpose tool. The excavator body is either mounted on a tracked or wheeled lower travel carriage.

4.1.2.1.2. Role in the mission (Unit/Base/Community (during natural disasters)/Air Force).

4.1.2.2. Understand the importance of efficient operation and performance of preventative maintenance on the excavator to meet mission requirements. Preventative maintenance ensures safe operation and availability for daily and emergency use.

4.1.2.3. Understand the safety precautions to be followed pre-, during- and post-operational inspection of the excavator.

4.1.2.4. Be completely familiar with the safety features of the excavator.

4.1.2.5. Safely and proficiently operate the excavator.

4.1.3. Excavator design. The design of an excavator varies depending on the vehicle manufacturer. Refer to the manufacturer's operator's manual(s) for additional information on the specific excavator being operated.

4.1.4. The operator must know the location and function of all controls and indicators prior to operating the vehicle.

Table 4.1. Controls and Indicators.

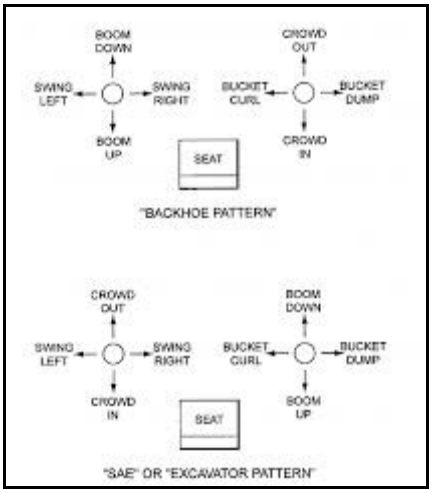
Control/Indicator	**Description**
Front instrument panel indicators	Displays all warning indicators and engine control information
Engine coolant gauge	Indicates coolant temperature
Fuel gage	Indicates the fuel level
Engine oil pressure warning indicator	Lights when engine oil pressure is too low
Engine coolant temperature warning indicator	Lights when the coolant temperature gets too high
Coolant level indicator light	Lights when the coolant level is too low
Air cleaner clogging warning indicator	Lights when the air cleaner is clogged
Air preheating indicator	Lights when the preheating is activated
Battery charge warning indicator	Lights when there are malfunctions in the charging system
(If applicable) Quick-coupling indicator	Lights when the quick-coupling is opened, provided that the two quick coupling switches on the left instrument panel and the right wall panel are pressed simultaneously
Caution: If the red warning lamp for the open quick-coupling lights up on the front instrument panel, while the bucket is still attached to the bracket, the excavator equipment should not be operated (as the bucket may suddenly loosen).	
(If applicable) Overload warning indicator	Lights if the attachment becomes overloaded, provided that the switch on the right instrument panel is pressed down

Caution: If the overload warning lamp lights up, stop the lifting operation and reduce the load. Otherwise, this can cause serious accident.	
Boost indicator	Lights when the boost power function has been selected with the right control lever
(If applicable) Float operation indicator	Lights when the float function has been selected with the right lever
(If applicable) Hammer indicator	Lights when the hammer button on the right control lever is pressed while hammer of boost/hammer switch on the right instrument is selected

(If applicable) Shear selecting indicator	Lights when the shear switch on the right instrument panel has been pressed
Switches and controls	**Note:** Refer to operator's manual
Engine speed control switch	Used to select the engine speed
Work lights switch	Controls the working lights
Power maximum mode selector switch	Used to activate the power maximum mode
Auto idle selector	Used to return the engine to idle when there is no load demanded
(If applicable) Boost/hammer/shear selector switch	Used to operate either the boost mode, hammer mode or shear mode
(If applicable) Right quick-couple switch	Used when the left quick-couple unit is opened.
Note: This function is activated by pressing the quick-couple switch on the left and right instrument panels simultaneously.	
(If applicable) Quick fit audible warning switch	Used to operate quick-couple warning buzzer.
Wiper switch	Controls upper/lower wipers
Washer switch	Controls windshield washers
Caution: Never press the washer switch for more than 20 seconds. Do NOT use if the washer fluid container is empty.	
Starter switch	Used to start the excavator
(If applicable) Extra work lamp switch	Controls the work lamps on the cab and counter weight
Beacon light switch	Controls the beacon light
Cab dome light switch	Turns dome light on and off
(If applicable) Seat heating switch	Used to turn the seat heater on/off
(If applicable) Overload warning switch	Controls the warning lamp and buzzer
Caution: When the overload warning lamp is on, stop the lifting operation and reduce the load. If not, serious and even fatal accidents can occur.	
(If applicable) Left quick-coupling switch	Used when the left quick-coupling unit is open
Note: This function is activated by pressing the quick-couple switch on the left and right instrument panels simultaneously.	
Travel warning sound stop switch	Used to mute the travel warning sound
Hour meter indicator	Displays how many hours the engine has run
(If applicable) One/two pump selector switch	Used when the boost/hammer/shear switch is set to hammer or shear
Automatic/manual selector switch	Used if the V-ECU malfunctions; the switch must be placed into the manual position
(If applicable) Engine diesel heater control	Controls the diesel heater. Refer to the operator's manual
Service socket	Used to connect diagnostic equipment

A/C and heater switch	Used to turn the A/C and heater on/off
Left operating lever	Controls arm out/arm in functions
	Controls left/right swing of the excavator
	Rotator buttons
	Horn buttons
Right operating lever	Controls bucket in/out functions
	Controls boom raise/lower functions
	Boost/hammer button
	Horn button
Travel levers and pedals (for tracked excavators)	
Caution: Keep feet clear of the travel pedals and levers when working. Know the track direction before operating travel pedals and levers. Travel operation will be reversed when the sprocket is at the front.	
Center position neutral	
Forward position of both pedals/levers move the excavator forward	
Rear position of both pedals/levers move the excavator in reverse	
(If applicable) Optional pedal left hand side	(adjustable boom)
(If applicable) Optional pedal right hand side	Operates hammer/shear or crusher

Figure 4.1. Control Pattern.



4.2. Vehicle Inspection.

4.2.1. Pre-operation vehicle inspection test. Use DAF Form 1800 as a 360 walk-around guide.

4.2.2. A Seven-Step Inspection Method will help ensure the inspection is the same each time it is conducted, and that nothing is left out. See **Attachment 3** for the Seven-Step Inspection Method.

4.2.3. Types of Vehicle Inspection. If discrepancies are found the operator must report them to VCO, the supervisor, and/or vehicle maintenance:

4.2.3.1. Pre-operation inspection – identify items/problems that could cause accidents or breakdowns.

4.2.3.1.1. Vehicle Maintenance may authorize continued use for all other maintenance discrepancies.

4.2.3.1.2. Cleanliness/damaged/missing items.

4.2.3.1.3. Leaks (fuel/oil/coolant/air).

4.2.3.1.4. Fluid levels; ensure levels are within limits:

4.2.3.1.4.1. Engine oil.

4.2.3.1.4.2. Coolant.

4.2.3.1.4.3. Power steering fluid.

4.2.3.1.4.4. Transmission fluid.

4.2.3.1.4.4. Hydraulic fluid.

Figure 4.2. Hydraulic Fluid Indicator.



4.2.3.1.5. Battery; security, fluid, damage and corrosion.

4.2.3.1.6. All wheel rims (cracks, splits, etc.); check for loose or missing lug nuts.

4.2.3.1.7. All tires.

4.2.3.1.7.1. Proper inflation.

4.2.3.1.7.2. Sidewalls, tread, to include depth, bulges.

4.2.3.1.7.3. Cuts and abrasions.

4.2.3.1.7.4. Lug nuts.

4.2.3.1.8. Transmission.

4.2.3.1.8.1. Differential(s). Damage, wear and leaks.

4.2.3.1.8.2. Drive train. Damage, wear and leaks.

4.2.3.1.9. Drive belts; tension and fraying.

4.2.3.1.10. Air filter(s).

Figure 4.3. Engine Compartment.



4.2.3.1.11. All hoses and wiring.

4.2.3.1.12. Suspension.

4.2.3.1.12.1. Shocks and springs, damage.

4.2.3.1.13. Frame bolts and other fasteners, visual inspection for damage.

4.2.3.1.14. Welds visual inspection for cracks.

4.2.3.1.15. Visual and auditory warning devices.

4.2.3.1.16. Storage bin doors properly latched, if applicable.

4.2.3.1.17. Pintle hook connection/compatibility, if applicable.

4.2.3.1.18. Fuel tank(s) assembly for damage.

4.2.3.1.19. Diesel exhaust fluid (DEF) tank, if applicable.

4.2.3.1.20. Wiring/lights/reflectors (interior/exterior).

4.2.3.1.21. Mirrors.

4.2.3.1.22. Windshield and windshield wipers/washers.

4.2.3.1.23. Doors.

4.2.3.1.24. Windows.

4.2.3.1.25. Seatbelts.

4.2.3.2. During-operation inspection.

4.2.3.2.1. Ensure master switch is turned to the ON position.

Figure 4.4. Master Switch.



4.2.3.2.2. Ignition to accessory position.

4.2.3.2.3. Check all gauges and warning lights/indicators for proper operations.

4.2.3.2.3.1. CAUTION: Regeneration system. Refer to technical manual(s).

4.2.3.2.4. Ignition to start.

4.2.3.2.5. Check for unusual conditions (interior).

4.2.3.2.5.1. Sounds.

4.2.3.2.5.2. Odors.

4.2.3.2.5.3. Vibrations.

4.2.3.2.6. Conduct 360 walk-around; check for unusual conditions (exterior).

4.2.3.2.6.1. Sounds.

4.2.3.2.6.2. Odors.

4.2.3.2.6.3. Vibrations.

4.2.3.2.6.4. Leaks.

4.2.3.2.6.5. Light function.

4.2.3.2.7. Conduct function check of all controls.

Figure 4.5. Steering Wheel (Wheeled model)



Figure 4.6. Track lever/pedal Controls (Tracked Model)



Figure 4.7. Operator Control (right).



Figure 4.8. Operator Controls (Left)



Figure 4.9. Safety Lever.



4.2.3.2.7.1. Steering wheel. **Note:** Disengage all-wheel steer function.

4.2.3.2.7.2. Track level/pedals

4.2.3.2.7.3. Safety selector.

4.2.3.2.7.4. Parking brake

4.2.3.2.7.5. Windshield wipers.

4.2.3.2.7.6. Climate control.

4.2.3.2.7.7. Conduct 360 walk-around; check for unusual conditions (exterior).

4.2.3.2.7.7.1. Sounds.

4.2.3.2.7.7.2. Odors.

4.2.3.2.7.7.3. Vibrations.

4.2.3.2.7.7.4. Leaks.

4.2.3.2.8. Sign DAF Form 1800. Verify Standard Form (SF) 91, Motor Vehicle Accident Report, SF 94, *Statement of Witness*, and Department of Defense (DD) Form 518, *Accident Identification Card are on-hand*.

4.2.3.2.9. Post-operation inspection.

4.2.3.2.10. Check fuel level (< ¾ tank, refuel).

4.2.3.2.11. Check DEF level (< ¾ tank, refuel), if applicable.

4.2.3.2.12. Ensure vehicle and components are cleaned.

4.2.3.2.13. Park vehicle. Ensure transmission in neutral, apply parking brake (if equipped).

4.2.3.2.14. Follow manufacturer's shut-down procedures.

4.2.3.2.15. Shut off lights and accessories.

4.2.3.2.16. Ensure master switch is turned to the OFF position

4.2.3.2.17. Post 360 walk-around

4.3. Vehicle Safety and Equipment.

4.3.1. Hazards and human factors:

4.3.1.1. Traffic due to size and weight.

4.3.1.2. Jerky starts and stops.

4.3.1.3. Traveling too fast and turning too sharply.

4.3.1.4. Slip hazards.

4.3.1.4.1. Always maintain three-points of contact when mounting/dismounting the vehicle.

4.3.1.5. High rollover risk.

4.3.1.6. Restricted visibility.

4.3.2. Safety clothing and equipment:

4.3.2.1. Safety-toed boots must be worn.

4.3.2.2. Leather gloves.

4.3.2.3. Hearing protection.

4.3.2.4. Inclement weather gear, if required.

4.3.2.5. Reflective belt during hours of reduced visibility and on flightline.

4.3.2.6. First aid kit.

4.3.2.7. Cones.

4.3.2.8. Tire gauge.

4.3.2.9. Fire extinguisher.

4.3.2.10. DAF Form 1800, SF 91 and DD Form 518.

4.4. Driving Safety and Precautions.

4.4.1. Overview safety and precautions. The following are safety items and procedures to be followed during excavator operations. The manufacturer's operator's manual will also provide safe operating procedures and the vehicle itself may have warnings, cautions and danger stickers that the vehicle operator should be aware of.

4.4.2. Vehicle data plate. Be familiar with the location and information found on the data plate.

4.4.3. Plan the route.

4.4.3.1. Overhead clearance. Check the clearance height of the vehicle relative to the overhead obstructions such as power lines, trees, and bridges. Refer to DAFMAN 91-203 for guidance on working in areas with overhead power lines and the distance the operator must maintain.

4.4.3.2. Width restrictions/construction zones, over-the-road.

4.4.3.3. Weight restriction (roads, bridges, off-road conditions).

4.4.3.4. Inclines.

4.4.3.5. Uneven ground.

4.4.3.6. Soft surfaces.

4.4.4. Over the road operation.

4.4.4.1. Greater vehicle weight. The operator needs to consider the combined weight of the excavator and the load. This will affect the following:

4.4.4.1.1. Operator's ability to stop. Do not tailgate the vehicle in front. Allow more distance between vehicles in order to increase reaction time.

4.4.4.1.2. Vehicle's ability to accelerate/follow the flow of traffic. Accelerate smoothly and gradually so the vehicle does not jerk. Rough acceleration causes unnecessary, premature mechanical damage to the vehicle's drive train. Maintain a safe speed.

4.4.4.2. Downgrades/upgrades. The operator will use lower gears more frequently to climb hills or mountains with increasing grade steepness, length and/or heavy load weight. Plan ahead to identify downgrades/upgrades on the route of travel. If possible, talk to other drivers who are familiar with the grades to find out what speeds are safe. When encountering downgrades/upgrades as described, the operator will need to address:

4.4.4.2.1. Speed. On downgrades, gravity causes the speed of the vehicle to increase. The operator must select an appropriate safe speed, use a low gear, and proper braking techniques. The operator must go slow enough so as to not overheat the vehicle's brakes.

4.4.4.2.2. Stopping. If the brakes become too hot, they may start to "fade". Brake fade will cause partial or complete loss of the brakes.

4.4.4.3. Sharp turns. Slow down before entering the turn. During the turn, avoid sharp sudden movements with the steering wheel. This reduces the chance of the vehicle weight shifting, and also prevents the possibility of tipping over due to the higher center of gravity.

4.4.4.4. Surroundings. Operating a excavator requires the operator's constant attention. Many situations can be avoided by simply paying close attention to the surrounding conditions. Road signs such as "steep grade", "low overhead clearance", "sharp turn ahead", and special speed limits are posted for the driver's safety.

4.4.4.5. Blind spots. Operators must know where there will be limited or no visibility surrounding the vehicle being operated.

4.4.4.6. Size. The operator must take into account, the size/width of the attachment assembly when operating the vehicle.

4.4.5. Backing.

4.4.5.1. Use a spotter and hand signals.

4.4.5.2. Back slowly and keep the spotter in view at all times. If the operator loses sight of the spotter, the operator must immediately stop the vehicle.

4.4.5.3. See QTP 24-3- General Vehicle Safety for additional guidance on spotter safety and for DAF standard spotter hand signals.

4.4.6. General operation.

4.4.6.1. Walk around the vehicle to ensure the area is clear before moving.

4.4.6.2. Before operating, the operator must understand all controls. He/she should ASK, if they do not understand!

4.4.6.3. Never attempt to start or operate the vehicle from any location other than the operator's seat.

4.4.6.4. Never leave the vehicle running unattended.

4.4.6.5. Do not attempt to get on or off of the machine while it is moving.

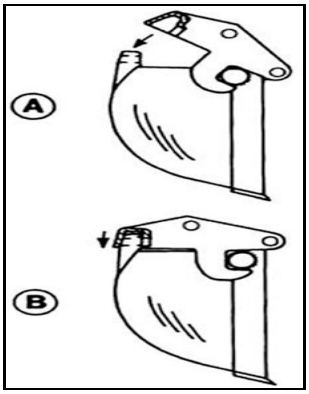
4.5. Vehicle Operation.

4.5.1. Attachment installation/removal. Different types of excavator attachments such as the impactor, compactor and bucket can greatly increase the versatility of the machine.

4.5.1.1. CAUTION: Before you install/remove an attachment, be sure the excavator is resting securely on solid level ground. To install an attachment on a typical excavator.

4.5.1.2. Attachment installation.

Figure 4.10. Attachment Installation.



4.5.1.2.1. Insert the hook onto the cross member of the attachment (**Figure 4.9.** – A).

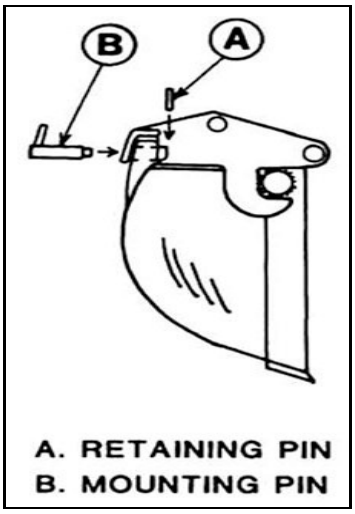
4.5.1.2.2. Extend the bucket cylinder until the rear connector engages the aligning bearing (**Figure 4.9.** – B).

4.5.1.2.3. Raise the attachment completely off the ground. Insert the mounting pin into the quick coupler, and install the retaining pin back into the mounting pin.

4.5.1.2.4. Depending on the attachment, connect the two hydraulic hoses.

4.5.1.3. Attachment removal.

Figure 4.11. Attachment Removal.



4.5.1.3.1. Depending on the attachment, disconnect the two hydraulic hoses.

4.5.1.3.2. Remove the retaining pin from the mounting pin.

4.5.1.3.3. Remove the mounting pin from the quick coupler (**Figure 4.10.**).

4.5.1.3.4. After you remove the mounting pin, take pressure off the bucket. Retract the bucket cylinder and remove the attachment.

4.5.2. General vehicle operations.

4.5.2.1. Complete a pre-operation vehicle inspection.

4.5.2.2. Sign the current DAF Form 1800.

4.5.2.3. Climb into the vehicle. Use three points of contact.

4.5.2.4. Adjust the seat and mirrors as needed; fasten seat belt.

4.5.3. Starting the vehicle.

4.5.3.1. Turn on the master switch.

4.5.3.2. Place safety locking system handle in the down position.

4.5.3.3. Make sure the operation control lever is in the disengaged position.

4.5.3.4. With the gear selector in neutral and the parking brake applied, turn the ignition switch to the start position. When the engine starts, release the ignition switch.

4.5.3.4.1. CAUTION: Do not engage the starter for more than 30 seconds at a time. If the engine does not start within 30 seconds, allow the starter two minutes to cool-off.

4.5.3.4. After the engine starts, ensure that oil pressure is indicated on the gauge within 15 seconds after starting. Idle engine for 3 to 5 minutes at 1000 Revolutions Per Minute (RPM) before operating.

4.5.4. Operating procedures (working with the bucket). The excavator is a multi-task machine capable of being fitted with a multitude of special attachments to perform many types of work. Only the simplest operations are described below. **Note:** You must have an approved DAF Form 103, *Base Civil Engineering Work Clearance Request*.

4.5.5. Backhoe work.

4.5.5.1. For digging work at a lower level than the machine is located:

4.5.5.1.1. When the angle between the bucket cylinder and links, arm cylinder and arm is set to 90 degrees, respectively, the working efficiency of cylinder will be at its maximum.

4.5.5.1.2. In case of digging, take advantage of this angle to improve the work efficiency.

4.5.5.1.3. When digging the useful movement of the arm is 30 degrees forward and 45 degrees rearward. There may be a little difference according to digging depth.

4.5.5.1.4. Do not use the cylinder up to its stroke end, but only within this range. Machine position is critical when digging large excavations, as seen in **Figure 4.4.**

4.5.6. Ditching work.

4.5.6.1. Install the proper bucket for ditching.

4.5.6.2. Place the machine over where the ditch is to work effectively.

4.5.6.3. In the case of a wide ditch, dig both sides first, then the center area.

Figure 4.12. Ditching Work.



4.5.7. Loading work. Loading a dump truck with an excavator is different than loading with other equipment (loaders, backhoes, etc.).

4.5.7.1. The excavator is normally positioned on top of the stockpile above and behind the dump truck. See **Figure 4.12.**

4.5.7.2. Position the hauler so as to achieve a small slewing movement and good visibility for the operator to work effectively.

4.5.7.3. Also, load over the rear of the dump truck, rather than the side. This makes it easier for the excavator to swing the machine and place the material in the dump. This also prevents the possibility of material being dropped on the truck cab.

4.5.8. When working with the bucket:

4.5.8.1. Do not strain the slewing mechanism. Do not use the slewing force for raking over the ground, demolition of buildings or thrusting bucket teeth in the ground. This operation may cause damage to the excavator and its attachments.

4.5.8.2. Do not work with the traveling motors. Do not dig by using the traveling motors and thrusting the bucket teeth into the ground. This can overload the rear of the excavator and damage the track drive.

4.5.8.3. Do not extend the hydraulic cylinder to its end of stroke. This can overload the stop in the cylinder and shorten the life span of the excavator. Work with as much clearance as possible.

4.5.8.4. Do not work by slamming the bucket into the ground. Do not perform digging by dropping the boom or using the bucket instead of a pick. Striking, digging or deliberate striking can overload the rear of the excavator or damage the attachment.

4.5.8.5. Do not carry out lifting work. Using the excavator as a crane is prohibited. Know the municipal, provincial, state and/or national regulations governing lifting work.

4.5.8.6. If lifting work is permitted, a properly installed rated bucket hook and certified slings/shackles are required.

4.5.8.7. Do not drop with the force of the excavator body. Do not operate by dropping the excavator body.

Figure 4.13. Dump Truck Loading.



4.5.9. Working on a slope.

4.5.9.1. While traveling on a slope, keep the angle between the boom and arm at 90-110 degrees. Raise the bucket 20-30 cm from the ground.

4.5.9.2. Do not descend backwards on a slope.

4.5.9.3. Do not change the direction or travel across on a slope. Change the direction on level ground. If necessary, first come down to level ground and make a detour.

4.5.9.4. If the excavator slides, immediately lower the bucket to the ground.

4.5.9.5. Do not perform swing work or operation of attachments. The machine can turn over if unbalanced. Do not swing a loaded bucket. In an unavoidable case, pile up earth on the slope and make the excavator level and stable.

4.5.9.6. Do not travel on a slope of 30 degrees or more.

4.5.9.6.1. CAUTION: Be cautious when operating on a slope. If the engine shuts down on a slope, do not operate the swing function. The superstructure may be swung under its own weight and cause tipping or side slipping. Be careful when opening/closing doors. Operational force may change rapidly. Keep doors closed when operating and when the vehicle is not in use.

4.5.9.7. When operating up a slope of 15 degrees or more, position the excavator so it is square to the slope. Keep the angle between the boom and arm at 90 to 110 degrees. Raise the bucket 20 to 30 cm from the ground.

4.5.9.8. If the track shoe slips on a slope, thrust the bucket into the ground. Pull the arm in to assist the track drive to move the excavator up the slope.

4.5.9.9. When operating down a slope of 15 degrees or more, position the excavator so it is square to the slope. Keep the angle between the boom and arm at 90 to 110 degrees. Raise the bucket 20 to 30 cm from the ground and travel at a slow speed.

Figure 4.14. Working on a Slope.



4.5.10. Operation of the hammer attachment.

4.5.10.1. The main uses of the hammer attachment are breaking stone, demolition work and road repair. It is widely used for demolition of buildings, breaking road surfaces, tunnel work, smashing slag and breaking or cutting stone.

4.5.10.2. Press the chisel firmly onto the surface at the right angle.

4.5.10.3. When striking, press the chisel firmly onto the surface. Lift the frame about 5 cm.

4.5.10.3.1. CAUTION: Never raise the excavator unnecessarily high.

4.5.10.4. If the surface is struck repeatedly, but it does not break within one minute, move the chisel nearer one end and strike again.

4.5.10.5. The striking direction of the chisel and the direction of the breaker body are deviate slightly. Adjust the bucket cylinder so that the direction of body and chisel is always the same.

4.5.10.6. Press the chisel firmly against the surface to avoid idle striking.

4.5.10.7. During hammer operation:

4.5.10.7.1. Do not operate the cylinder to its end of stroke; leave approximately 5 cm.

4.5.10.7.2. Do not swing the hammer against the rocks, concrete, etc.

4.5.10.7.3. Do not move the chisel while it is striking a blow.

4.5.10.7.4. Do not strike vertically or in an upward direction.

4.5.10.7.5. Do not bend the chisel to make a hole in the ground.

4.5.10.7.6. Do not hoe the chisel.

4.5.10.7.7. Do not raise the excavator by extending the bucket cylinder to its maximum.

4.5.11. Shutdown procedures.

4.5.11.1. Bring the excavator to a complete stop.

4.5.11.2. Return the travel levers/pedals to the neutral position.

4.5.11.3. (If applicable) Apply parking brake.

4.5.11.4. Lower all attachments to the ground.

4.5.11.5. Turn the engine speed control switch to idle.

4.5.11.6. Allow engine three to five minutes to cool down.

4.5.11.7. Turn the key to the OFF position.

4.5.11.8. Check for damage.

4.5.12. End of duty day.

4.5.12.1. Perform post-operation procedures as described in **Paragraph 4.2.**

4.5.12.2. Cleaning air intake filters. There are generally two elements, the inner and the outer. Under dusty operating conditions, clean outer elements daily (even more often if working conditions are extremely dusty). The inner filter will be replaced during regular scheduled maintenance. For cleaning procedures, use guidelines stated in the operator's maintenance manual.

4.5.12.3. Lubricating the vehicle according to intervals listed in the maintenance chart. If operating the machine in severe conditions, lubricate the machine more frequently.

4.5.12.4. Fuel the Excavator at the end of each working day to prevent moisture from condensing and forming droplets of water within the fuel tank. Contact base fuels to come to the job site if your equipment can't be driven to the service station (i.e., extreme distances, tracked vehicles, no drivable support equipment, etc.). Ensure the vehicle has a minimum of three-fourths tank of fuel at the end of the duty day.

Chapter 5

EXPLANATION AND DEMONSTRATION

5.1. Instructor's Preparation.

5.1.1. Establish a training location.

5.1.2. Obtain appropriate vehicle operator's manual.

5.1.3. Schedule/reserve a vehicle.

5.1.4. Ensure trainee completes DAF Form 171.

5.2. Safety Procedures and Equipment.

5.2.1. The following safety items should be followed by both the instructor and trainee.

5.2.1.1. Chock wheel (if required) when excavator is parked.

5.2.1.2. Remove all jewelry and identification tags.

5.2.1.3. Personal protective equipment (PPE) and equipment items.

5.2.1.3.1. Safety toed boots must be worn.

5.2.1.3.2. Gloves will be worn during pre-operation inspection, post-operation inspection and while performing maintenance/adjustments to the attachment.

5.2.1.3.3. Hearing protection, if required

5.2.1.3.4. Eye protection, if required.

5.2.1.3.5. Inclement weather gear.

5.2.1.3.6. Reflective belt during hours of reduced visibility or on the flightline.

5.2.1.3.7. Warning triangles.

5.2.1.4. The trainer and the trainee should conduct a 360 walk-around the vehicle to become familiar with all warning labels and signs.

5.2.1.5. Ensure that the vehicle is properly parked and the brakes are set before accomplishing the walk-around inspection.

5.2.1.6. Properly adjust driver's seat and all mirrors.

5.2.1.7. Ensure trainee wears seat belt.

5.2.1.8. Throughout demonstration, practice excavator operational safety.

5.2.2. Practice basic AF RM process during demonstration:

5.2.2.1. Identify the hazards.

5.2.2.2. Assess the hazards.

5.2.2.3. Develop controls and make decisions.

5.2.2.4. Implement controls.

5.2.2.5. Supervise and evaluate.

5.3. Operator Maintenance Demonstration.

5.3.1. With trainee, accomplish vehicle inspection using DAF Form 1800. The vehicle inspection will follow the seven-step method as described in **Attachment 3**. An inspection guide can be used to ensure all areas of the excavator are covered in addition to the "Operation Demonstration" guidelines provided below.

5.4. Operation Demonstration.

5.4.1. Throughout demonstration:

5.4.1.1. Allow for questions.

5.4.1.2. Repeat demonstrations as needed.

5.4.2. Demonstrate/discuss pre-operation and during-operation inspection requirements.

5.4.3. Describe the operation and location of the following items:

5.4.3.1. Knob throttle.

5.4.3.2. Steering controls.

5.4.3.3. Horn button.

5.4.3.4. Gear selector.

5.4.3.5. Hydraulic controls and functions.

5.4.3.6. Parking brake.

5.4.3.7. Operation safety lock control lever.

5.4.4. Discuss the following important operational notes:

5.4.4.1. Radiator checks. When cold, the coolant level should be approximately 1 inch from the top of the filler neck and the full cold mark on the reservoir.

5.4.4.2. Typically, transmission fluid must be checked with the transmission warm, engine running and gear selector in the neutral position. Excavator transmissions vary from model to model and may require different fluid checking procedures. Check the operator's manual prior to checking the fluid.

5.4.4.3. Engine oil must be at the full mark on the oil dipstick.

5.4.4.4. Do not operate the starter for more than 30 seconds. If the engine does not start within 30 seconds, allow the starter motor to cool for 2 minutes before attempting to restart the engine.

5.4.4.5. If a wheel is removed and replaced for puncture repair or any other reason, the wheel nuts must be tightened to the torque specification (ft/lbs) provided in the operator's manual and checked daily until stabilized.

5.4.5. Demonstrate the following for the excavator.

5.4.5.1. Proper mounting and dismounting procedures.

5.4.5.2. Engine start up, including proper safety precautions.

5.4.5.3. Instrument use and their indications.

5.4.5.4. Proper use of excavator controls.

5.4.5.5. Proper movement with and without a load.

5.4.5.5.1. Forward.

5.4.5.5.2. Turning. (at various speeds)

5.4.5.5.3. Braking.

5.4.5.5.4. Backing, (use spotter when backing).

5.4.5.5.5. Parking.

5.4.5.6. Excavator operations. **Note:** Refer to the technical manual for additional guidance pertaining to the vehicle being operated. Demonstrate:

5.4.5.6.1. Backhoe work.

5.4.5.6.2. Ditching work.

5.4.5.6.3. Loading work.

5.4.5.6.4. Working on a slope.

5.4.5.6.5. Changing attachments.

5.4.5.6.6. Operating the hammer.

5.4.5.6.7. Demonstrate shutdown procedures.

5.4.6. Demonstrate/discuss post-operation requirements.

5.4.6.1. Ensure vehicle is clean.

5.4.6.2. Refuel vehicle.

5.4.6.3. Following manufacturer's shut-down procedures.

5.4.6.4. Perform a 360 walk-around inspection.

5.4.6.5. Annotate any discrepancies found on DAF Form 1800.

5.4.7. Conclude by allowing time for questions and any requested re-demonstrations.

Chapter 6

TRAINEE PERFORMANCE AND EVALUATION

6.1. Trainee Performance.

6.1.1. Instructor will:

6.1.1.1. Ensure safety at all times. **Note:** Stop training when safety items are violated. Proceed only when the trainee fully understands how to avoid repeating the safety infraction(s).

6.1.1.1.1. Chock wheel (if required) when excavator is parked.

6.1.1.1.2. Remove all jewelry and identification tags.

6.1.1.1.3. If available, mark vehicle with magnetic sign indicating “Driver-in-Training” or “Trainee Operator.”

6.1.1.2. PPE and other items: **Note:** Discuss when it is required that applicable PPE should be worn/utilized.

6.1.1.2.1. Safety toed boots must be worn.

6.1.1.2.2. Gloves will be worn during pre-operation, post-operation inspection and while performing maintenance/adjustments to the attachment.

6.1.1.2.3. Hearing protection, if required.

6.1.1.2.4. Eye protection, if required.

6.1.1.2.5. Reflective belt during hours of reduced visibility or on the flightline

6.1.1.2.6. Warning triangles.

6.1.1.2.7. Inclement weather gear, if required.

6.1.1.3. Pay particular attention to the cautions and warnings listed in the operator's manual.

6.1.1.4. Properly adjust driver’s seat and all mirrors.

6.1.1.5. Ensure trainee wears seat belt.

6.1.1.6. Excavator safety items/procedures.

6.1.1.7. Ensure the trainee is aware of tasks to be performed.

6.1.1.8. Conduct during/after-action reviews with the trainee. (Demonstration may need to be re-accomplished).

6.1.2. Trainee Performance.

6.1.2.1. Conduct operator maintenance (have trainee explain items being inspected).

6.1.2.1.1. Pre-operation inspection.

6.1.2.1.2. During-operation inspection.

6.1.2.2. Ensure DAF From 1800 is properly documented.

6.1.2.2.1. Identify and explain excavator gauges, switches, levers and buttons.

6.1.2.2.2. Establish a road course that will have the following: (if the course does not have one of the following, then the trainee should be able to explain the correct operating techniques).

6.1.2.2.2.1. Forward.

6.1.2.2.2.2. Turning.

6.1.2.2.2.3. Braking.

6.1.2.2.2.4. Backing (use spotter when backing).

6.1.2.2.2.5. Parking.

6.1.2.2.2.6. Excavator operation. **Note:** If the course does not have one of the following, then the trainee should be able to explain the correct operating techniques

6.1.2.2.2.6.1. Backhoe work.

6.1.2.2.2.6.2. Ditching work.

6.1.2.2.2.6.3. Loading work.

6.1.2.2.2.6.4. Working on a slope.

6.1.2.2.2.6.5. Changing attachments.

6.1.2.2.2.6.6. Operating the hammer.

6.1.2.2.2.6.7. Shutdown procedures.

6.1.2.2.2.6.8. Refer to the technical manual for additional guidance pertaining to the vehicle being operated.

6.1.2.2.3. Perform post-operation inspection.

6.1.2.2.3.1. Ensure vehicle components are cleaned.

6.1.2.2.3.2. Check fuel level. If there is < ¾ tank, refuel the vehicle.

6.1.2.2.3.3. Check diesel exhaust fluid level, if equipped.

6.1.2.2.3.4. Following manufacturer's shut-down procedures.

6.1.2.2.3.5. Park.

6.1.2.2.3.5.1. Place transmission in neutral.

6.1.2.2.3.5.2. Apply parking brake.

6.1.2.2.3.6. Perform a 360 walk-around inspection checking for leaks and damage.

6.2. Performance Evaluation.

6.2.1. Trainee will perform performance evaluation found in **Attachment 2**. **Note:** If available, mark vehicle with magnetic sign indicating "Driver-in-Training" or "Trainee Operator".

6.2.1.1. Instructor and trainee will review **Attachment 2**.

6.2.1.2. Instructor will answer trainee's questions.

6.2.2. Instructor will:

6.2.2.1. Ensure safety at all times.

6.2.2.1.1. Place wheel chocks (if required) when excavator is parked,

6.2.2.1.2. Remove all jewelry and identification tags.

6.2.2.2. PPE and other items.

6.2.2.2.1. Safety toed boots must be worn.

6.2.2.2.2. Gloves will be worn during pre-operation inspection, post- operation inspection and while performing maintenance/adjustments to the attachment.

6.2.2.2.3. Hearing protection, if required.

6.2.2.2.4. Eye protection, if required.

6.2.2.2.5. Reflective belt during hours of reduced visibility or on the flightline.

6.2.2.2.6. Warning triangles.

6.2.2.2.7. Inclement weather gear, if required.

6.2.2.3. Ensure trainee wears seat belt.

6.2.2.4. Properly adjust driver's seat and all mirrors.

6.2.2.5. Excavator safety items/procedures.

6.2.3. Explain operating techniques.

6.2.4. The trainee will demonstrate and be evaluated on the following procedures:

6.2.4.1. Vehicle/equipment checkout.

6.2.4.2. Pre-operation inspection/preventative maintenance.

6.2.4.3. Start-up procedures.

6.2.4.4. Forward.

6.2.4.5. Turning.

6.2.4.6. Braking.

6.2.4.7. Backing (use spotter when backing).

6.2.4.8. Parking.

6.2.4.9. Excavator operation. **Note:** Refer to the technical manual for additional guidance pertaining to the vehicle being operated.

6.2.4.9.1. Backhoe work.

6.2.4.9.2. Loading work.

6.2.4.9.3. Working on a slope.

6.2.4.9.4. Changing attachments.

6.2.4.9.5. Operating the hammer.

6.2.4.9.6. Shutdown procedures.

6.2.4.10. Perform post-operation inspection.

6.2.4.10.1. Ensure vehicle components are cleaned.

6.2.4.10.2. Check fuel level. If there is < ¾ tank, refuel the vehicle.

6.2.4.10.3. Following manufacturer's shut-down procedures.

6.2.4.10.4. Park.

6.2.4.10.4.1. Place transmission in neutral.

6.2.4.10.4.2. Apply parking brake.

6.2.4.10.5. Perform a 360 walk-around inspection checking for leaks and damage.

6.2.5. Ensure the driver is aware of operating situations.

6.2.6. Conduct after-action reviews with the trainee.

6.2.7. Trainee is not allowed any instructor assists to pass performance evaluation.

6.2.8. Evaluation checklist provided in **Attachment 2**.

6.2.9. Retraining; retrain No-Go's.

6.2.9.1. Re-demonstrate "No-Go" items.

6.2.9.2. Have trainee re-perform until they show proficiency in operating, critique weaknesses as observed.

6.2.9.3. Re-evaluate.

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DAFI 24-301, *Ground Transportation*, 09 January 2026

DAFI 24-302, *Vehicle Management*, 23 January 2026

DAFMAN 91-203, *Department of the Air Force Occupational Safety Fire and Health Standards,* 23 February 2026

DAFPAM 90-803, *Risk Management (RM) Guidance and Tools*, 16 February 2026

QTP24-3, *General Vehicle Safety*, 22 January 2026

Adopted Forms

DAF Form 103, *Base Civil Engineering Work Clearance Request,* 15 April 2025

DAF Form 171, *Request for Driver's Training and Addition to U.S. Government Drivers*, 05 May 2026

DAF Form 623, *Individual Training Record Folder,* 30 March 2023

DAF Form 847, *Recommendation for Change of Publication*, 16 April 2026

DAF Form 1800, *Operator's Inspection Guide and Trouble Report*, 17 February 2026

DD Form 518, *Accident Identification Card,* October 1978

SF 91, *Motor Vehicle Accident Report,* September 2020

SF 94, *Statement of Witness,* November 2020

Abbreviations and Acronyms

A/C—Air Conditioning

ABS—Anti-lock Braking System

AF—Air Force

AFIMSC—Air Force Installation Mission Support Center

DAF—Department of the Air Force

DAFI—Department of the Air Force Instruction

DAFMAN—Department of the Air Force Manual

DD—Department of Defense

DEF—Diesel Exhaust Fluid

IAW—In Accordance With

ITR—Individual Training Record

OTR—On-the-Job Training

PPE—Personal Protective Equipment

PSI—Pounds per Square Inch

QTP—Qualification Training Package

RM—Risk Management

RPM—Revolutions Per Minute

SF—Standard Form

UTM—Unit Training Manager

VCO—Vehicle Control Official

V-ECU—Vehicle Electronic Control Unit

Attachment 2

PERFORMANCE TEST

A2.1. Desired Learning Outcome.

A2.1.1. Understand the safety precautions to be followed pre-, during-, and post-operation of the excavator.

A2.1.2. Understand the purpose of the excavator and its role in the mission.

A2.1.3. Know the proper operator maintenance procedures of the excavator, IAW applicable technical orders and use of DAF Form 1800.

A2.1.4. Safely and proficiently operate the excavator.

A2.2. Instructions. Before beginning the performance test, the trainer will brief the trainee on the scenario that will need to be accomplished. He/she will be given additional directions and instructions as needed throughout the scenario.

A2.3. Scoring.

A2.3.1. The trainer examiner will be scoring the trainee on excavator operations and the general safe driving practices. The examiner will give directions and instructions to the trainee in sufficient time for him/her to conduct a driving maneuver. They will not be asked to drive in an unsafe manner.

A2.3.2. The examiner will be making various marks on the performance test checklist. This does not necessarily mean anything has been done wrong. It is in the best interest to concentrate on the operation of the excavator. The trainer will explain the test results at the conclusion of the performance test.

A2.3.3. Tasks being graded are listed on the following page; the trainee will be required to successfully pass all items.

A2.3.4. The instructor will stop the test at any time safe operations are not being followed or as deemed necessary for safety concerns.

Figure A2.1. Performance Test Checklist:

PERFORMANCE TEST CHECKLIST

Instructions to Certifier and Trainee:

Pre-Test Briefing: The Certifier will review this entire checklist with the trainee and brief the intended road course route. The trainee must understand that all driving directions will be provided in sufficient time to execute maneuvers safely.

Scoring Awareness: The Certifier will make various marks on this checklist during the evaluation. These marks do not inherently indicate a "No-Go"; the trainee must focus exclusively on safe vehicle operation.

Conduct of Test: No Trainer or Certifier assists are permitted during the evaluation. Results will be explained to the trainee upon the conclusion of the test.

Safety: The Certifier will stop the test immediately if unsafe operations occur or health concerns arise.

SECTION I. TRAINEE AND CERTIFIER INFORMATION

1.Trainee Name (Last, First MI)	**2. Unit/Office Symbol**
3. Certifier Name (Last, First MI)	**4. Unit/Office Symbol**

SECTION II. PERFORMANCE TEST INFORMATION

5. Vehicle Management Code for Vehicle Tested	**6. Date**

SECTION III. SCORESHEET

7. Equipment & Posture	Go	No Go
Verifies vehicle required forms & personal protective equipment	☐	☐

8. Pre and Post Operational checks & General Safety:	Go	No Go
Conducts 360 degree safety inspection when approaching vehicle	☐	☐
Inspects items identified on DAF1800, documents discrepancies	☐	☐
Conducts ops check of controls & instruments	☐	☐
Sign DAF1800	☐	☐
Performs post trip inspection,reports any discrepancies to Veh Mgt	☐	☐

9. On-Road Operational Compliance (Wheeled excavator):	Go	No Go
Obeys all traffic signs, signals, local/military traffic laws.	☐	☐
Completes course without accidents/moving violations.	☐	☐
Regular traffic/mirror checks/safe following distances.	☐	☐
Uses proper signals/smooth, safe lane changes.	☐	☐

10. Maneuvers:	Go	No Go
Turns: Checks traffic in all directions/signals properly.	☐	☐
Turns: Decelerates smoothly for the turn.	☐	☐
Turns: Checks mirrors/avoids drifting into oncoming traffic.	☐	☐
Curves: Reduces speed prior to entry.	☐	☐
Curves: Maintains lane positioning/360-degree traffic checks.	☐	☐

11. Intersections and Hazards (Wheeled excavator):	Go	No Go
Int.: Decelerates gently/brakes smoothly on approach.	☐	☐
Int: Stop (no coasting) behind stop lines/signs/sidewalks.	☐	☐
R/R: Decelerates and shifts gears before the crossing.	☐	☐
R/R: Does not stop, shift, or change lanes while crossing.	☐	☐
Signs: Identifies height clearance signs on route	☐	☐

12. General Vehicle Controls and Safety:	Go	No Go
Understands GVW, braking and stability.	☐	☐
Uses seat belts and all PPE	☐	☐
Verifies surroundings; uses mirrors and spotters; controls speed.	☐	☐
Maintains control of the boom and bucket or attachment.	☐	☐
Backing: Uses mirrors and spotter and voids blind side backing.	☐	☐

13. **Operations**:	Go	No Go
Manipulates controls smoothly without damage to equipment or surrounding.	☐	☐
Performs backhoe work with in reasonable accuracy.	☐	☐
Performs ditching work with in reasonable accuracy.	☐	☐
Loads material into a dump truck safely.	☐	☐
Operates on a slope safely.	☐	☐
Performs attachment change safely.	☐	☐
Operates hammer attachment safely.	☐	☐
Performs safe shutdown procedures and grounds bucket and/or attachments.	☐	☐

14. REMARKS

SECTION IV. CERTIFIERS ENDORSEMENT

I, the undersigned, do attest and certify that the trainee named herein has successfully completed all required training and is deemed proficient in the operation of the vehicle(s)/equipment listed in **Section II**. This certification is made to the best of my knowledge and belief, based on training conducted pursuant to the approved qualification training package.

14. Signature	**15. Date**

Attachment 3

SEVEN-STEP INSPECTION PROCESS

Figure A3.1. Seven-Step Inspection Process.

SEVEN-STEP VEHICLE INSPECTION GUIDE

Step 1: Vehicle Overview

Review Last Vehicle Inspection Report. Drivers may have to make a vehicle inspection report in writing each day. The motor carrier must repair any items in the report that affect safety and certify on the report that repairs were made or were unnecessary. Only sign the previous report after verifying that any noted defects have been repaired or certified as not needing repair..

Step 2: Check Engine Compartment

Parking Brakes Are On and/or Wheels Chocked.

You may have to raise the hood, tilt the cab (ssecure any loose items to prevent them from falling and causing damage), or open the engine compartment door.

Check the following:

- Engine oil level.
- Coolant level in radiator; condition of hoses.

Check That the

- Power steering fluid level; hose condition (if so equipped).
- Windshield washer fluid level

Battery fluid level, connections, and tie downs (battery may be located elsewhere)

- Automatic transmission fluid level (may require engine to be running).
- Check belts for tightness and excessive wear (alternator, water pump, air compressor) --learn how much "give" the belts should have when adjusted right and check each one.
- Leaks in the engine compartment (fuel, coolant, oil, power steering fluid, hydraulic fluid, battery fluid).
- Cracked, worn electrical wiring insulation.
- Lower and secure hood, cab, or engine compartment door.

Step 3: Start Engine and Inspect Inside the Cab

Get In and Start Engine

Make sure parking brake is on.

Put gearshift neutral (or "park" if automatic).

Start engine; listen for unusual noises.

If equipped, check the Anti-lock Braking System (ABS) indicator lights. Light on dash should come on and then turn off. If it stays on the ABS is not working properly. For trailers only, if the yellow light on the left rear of the trailer stays on, the ABS is not working properly.

Look at the Gauges

Oil pressure. Should come up to normal within seconds after engine is started. See Figure 2.5

Air pressure. Pressure should build from 50 to 90 psi within 3 minutes. Build air pressure to governor cut-out (usually around 120 – 140 psi. **Know your vehicle's requirements.**

Ammeter and/or voltmeter. Should be in normal range(s).

Coolant temperature. Should begin gradual rise to normal operating range.

Engine oil temperature. Should begin gradual rise to normal operating range.

Warning lights and buzzers. Oil, coolant, charging circuit warning, and antilock brake system lights should go out right away.

Check Condition of Controls. Check all of the following for looseness, sticking, damage, or improper setting:

- Steering wheel.
- Clutch.
- Accelerator ("gas pedal").
- Brake controls.
- Foot brake.
- Trailer brake (if vehicle has one).
- Parking brake.
- Retarder controls (if vehicle has them).
- Transmission controls.
- Interaxle differential lock (if vehicle has one).
- Horn(s).
- Windshield wiper/washer.
- Lights.
- Headlights.
- Dimmer switch.
- Turn signal.
- Four-way flashers.
- Parking, clearance, identification, marker switch(es).

OIL PRESSURE

- Idling 5-20 PSI
- Operating 35-75 PSI
- Low, Dropping, Fluctuating: **STOP IMMEDIATELY!** Without oil the engine can be destroyed rapidly

Figure 2.5

Check Mirrors and Windshield. Inspect mirrors and windshield for cracks, dirt, illegal stickers, or other obstructions to seeing clearly. Clean and adjust as necessary.

Check Emergency Equipment

Check for safety equipment:

Spare electrical fuses (unless vehicles have circuit breakers).

Three red reflective triangles, 6 fuses or 3 liquid burning flares.

Properly charged and rated fire extinguisher.

Check for optional items such as:

- Chains (where winter conditions require).
- Tire changing equipment.
- List of emergency phone numbers
- Accident reporting kit (packet).

Check Safety Belt. Check that the safety belt is securely mounted, adjusts; latches properly and is not ripped or frayed.

Step 4: Turn Off Engine and Check Lights

Make sure the parking brake is set, turn off the engine, and take the key with you. Turn on headlights (low beams) and four-way emergency flashers and get out of the vehicle.

Step 5: Do Walk-around Inspection

- Go to front of vehicle and check that low beams are on, and both of the four-way flashers are working.
- Push dimmer switch and check that high beams work.
- Turn off headlights and four-way emergency flashers.
- Turn on parking, clearance, side-marker, and identification lights.
- Turn on right turn signal and start walk-around inspection.

General

- Walk around and inspect.
- Clean all lights, reflectors, and glass as you go along.

Left Front Side

- Driver's door glass should be clean.
- Door latches or locks should work properly.
- Left front wheel.
- Condition of wheel and rim--missing, bent, broken studs, clamps, lugs, or any signs of misalignment.
- Condition of tires--properly inflated, valve stem and cap OK, no serious cuts, bulges, or tread wear.
- Use wrench to test rust-streaked lug nuts, indicating looseness.
- Hub oil level OK, no leaks.
- Left front suspension.
- Condition of spring, spring hangers, shackles, U-bolts.
- Shock absorber condition.
- Left front brake.
- Condition of brake drum or disc.
- Condition of hoses.

Front

- Condition of front axle and steering system.
- No loose, worn, bent, damaged or missing parts.
- Must grab steering mechanism to test for looseness.
- Condition of windshield.
- Check for damage and clean if dirty.
- Check windshield wiper arms for proper spring tension.
- Check wiper blades for damage, "stiff" rubber, and securement.
- Lights and reflectors.
- Parking, clearance, and identification lights clean, operating, and proper color (amber at front).
- Reflectors clean and proper color (amber at front).
- Right front turn signal light clean, operating, and proper color (amber or white on signals facing forward).

Right Side

- Right front: check all items as done on left front.
- Primary and secondary safety cab locks engaged (if cab-over-engine design).
- Right fuel tank(s).
- Securely mounted, not damaged, or leaking.
- Fuel crossover line secure.
- Tank(s) contain enough fuel.
- Cap(s) on and secure.
- Condition of visible parts.
- Rear of engine--not leaking.
- Transmission--not leaking.
- Exhaust system--secure, not leaking, not touching wires, fuel, or air-lines.
- Frame and cross members--no bends or cracks.
- Air-lines and electrical wiring--secured against snagging, rubbing, wearing.
- Spare tire carrier or rack not damaged (if so equipped).
- Spare tire and/or wheel securely mounted.
- Spare tire and wheel adequate (proper size, properly inflated).
- Cargo securement (trucks).
- Cargo properly blocked, braced, tied, chained, etc.
- Header board adequate, secure
- Side boards, stakes strong enough, free of damage, properly set in place
- Canvas or tarp properly secured to prevent tearing, billowing, or blocking of mirrors.
- If oversize, all required signs (flags, lamps, and reflectors) safely and properly mounted and all required permits in driver's possession.
- Curbside cargo compartment doors in good condition, securely closed, latched/locked and required security seals in place.

Right Rear

- Condition of wheels and rims--no missing, bent, or broken spacers, studs, clamps, or lugs.
- Condition of tires--properly inflated, valve stems and caps OK, no serious cuts, bulges, tread wear, tires not rubbing each other, and nothing stuck between them.
- Tires same type, e.g., not mixed radial and bias types.
- Tires evenly matched (same sizes).
- Wheel bearing/seals not leaking.
- Suspension.
- Condition of spring(s), spring hangers, shackles, and U-bolts.
- Axle secure.
- Powered axle(s) not leaking lube (gear oil).
- Condition of torque rod arms, bushings.
- Condition of shock absorber(s).
- If retractable axle equipped, check condition of lift mechanism. If air powered, check for leaks.
- Condition of air ride components.
- Brakes.
- Brake adjustment.
- Condition of brake drum(s) or discs.
- Condition of hoses--look for any wear due to rubbing.
- Lights and reflectors.
- Side-marker lights clean, operating, and proper color (red at rear, others amber).
- Side-marker reflectors clean and proper color (red at rear, others amber).

Rear

- Lights and reflectors.
- Rear clearance and identification lights clean, operating, and proper color (red at rear).
- Reflectors clean and proper color (red at rear).
- Taillights clean, operating, and proper color (red at rear).
- Right rear turn signal operating, and proper color (red, yellow, or amber at rear).
- License plate(s) present, clean, and secured.
- Splash guards present, not damaged, properly fastened, not dragging on ground, or rubbing tires.
- Cargo secure (trucks).
- Cargo properly blocked, braced, tied, chained, etc.
- Tailboards up and properly secured.
- End gates free of damage, properly secured in stake sockets.
- Canvas or tarp (if required) properly secured to prevent tearing, billowing, or blocking of either the rearview mirrors or rear lights.
- If over-length, or over-width, make sure all signs and/or additional lights/flags are safely and properly mounted and all required permits are in driver's possession.
- Rear doors closed, latched/locked.

Left Side

- Check all items as done on right side, plus:
- Battery(ies)
- Battery box(es) securely mounted.
- Box has secure cover.
- Battery(ies) secured against movement.
- Battery(ies) not broken or leaking.
- Fluid in battery(ies) at proper level (except maintenance-free type).
- Cell caps present and securely tightened (except maintenance-free type).
- Vents in cell caps free of foreign material (except maintenance-free type).

Step 6: Check Signal Lights

- Get In and Turn Off Lights
- Turn off all lights.
- Turn on stop lights (apply trailer hand brake or have a helper put on the brake pedal).
- Turn on left turn signal lights.
- Get Out and Check Lights
- Left front turn signal light clean, operating and proper color (amber or white on signals facing the front).
- Left rear turn signal light and both stop lights clean, operating, and proper color (red, yellow, or amber).

Get In Vehicle

- Turn off lights not needed for driving.
- Check for all required papers, trip manifests, permits, etc.
- Secure all loose articles in cab (they might interfere with operation of the controls or hit you in a crash).
- Start the engine.

Step 7: Start the Engine and Check

Test for Hydraulic Leaks. If the vehicle has hydraulic brakes, pump the brake pedal three times. Then apply firm pressure to the pedal and hold for five seconds. The pedal should not move.

If it does, there may be a leak or other problem. Get it fixed before driving. If the vehicle has air brakes, do the checks described in Sections 5 and 6 of this manual.

<u>Brake System</u>

Test Parking Brake(s)

- Fasten safety belt
- Set parking brake (power unit only).
- Release trailer parking brake (if applicable).
- Place vehicle into a low gear.
- Gently pull forward against parking brake to make sure the parking brake holds.
- Repeat the same steps for the trailer with trailer parking brake set and power unit parking brakes released (if applicable).
- If it doesn't hold vehicle, it is faulty; get it fixed.

Test Service Brake Stopping Action

- Go about five miles per hour.
- Push brake pedal firmly
- "Pulling" to one side or the other can mean brake trouble.
- Any unusual brake pedal "feel" or delayed stopping action can mean trouble.
- If any defects are found during the inspection, they must be repaired before operating the vehicle. Federal and state laws prohibit driving an unsafe vehicle.